

4.3.6

EE25BTECH11022 - sankeerthan

Question: passing through the points $(3, 4, -7)$ and $(1, -1, 6)$.

solution: let Passing through the points $\mathbf{A} = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix}$.

$$\begin{pmatrix} 3 & 4 & -7 \\ 1 & -1 & 6 \end{pmatrix} \mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Writing the augmented matrix:

$$\begin{pmatrix} 3 & 4 & -7 & 1 \\ 1 & -1 & 6 & 1 \end{pmatrix} \xleftrightarrow{R_1 \leftrightarrow R_1 - 3R_2} \begin{pmatrix} 0 & 7 & -25 & -2 \\ 1 & -1 & 6 & 1 \end{pmatrix} \xleftrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & -1 & 6 & 1 \\ 0 & 7 & -25 & -2 \end{pmatrix} \quad (1)$$

$$n_1 - n_2 + 6n_3 = 1 \quad (2)$$

$$7n_2 - 25n_3 = -2 \quad (3)$$

From third equation:

$$n_2 = \frac{-2 + 25n_3}{7} \quad (4)$$

Substitute in second equation:

$$n_1 = \frac{5 - 17n_3}{7} \quad (5)$$

Let $n_3 = t$, then

$$\mathbf{n} = \begin{pmatrix} \frac{5-17t}{7} \\ \frac{-2+25t}{7} \\ t \end{pmatrix} \quad (6)$$

let $t = 1$, multiply with 7 to clear denominators:

$$\mathbf{n} = \begin{pmatrix} -12 \\ 23 \\ 7 \end{pmatrix} \quad (7)$$

Thus, the equation of the line is: $(-12 \ 23 \ 7) \mathbf{x} = 7$

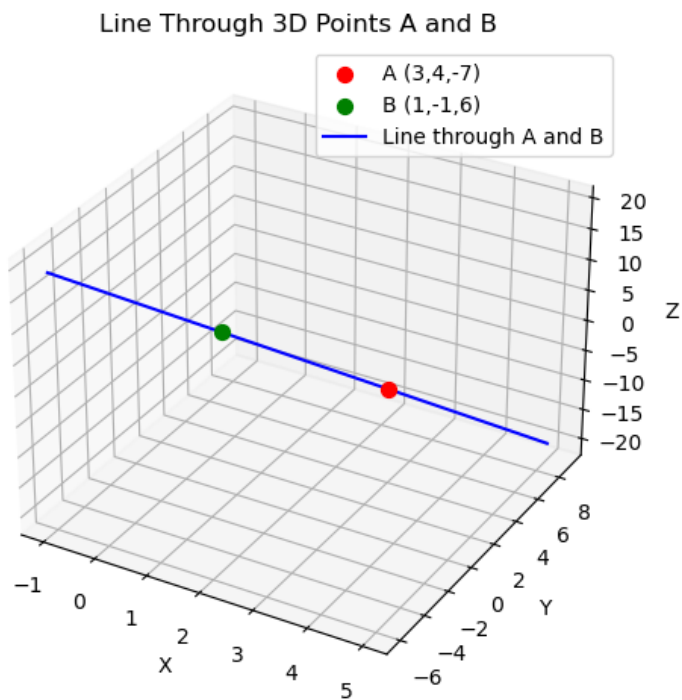


Fig. 0.